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Applied Calculus IV

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Week 11

Chapter 6



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Recall from calculus that power series in $x - a$ is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots$$

namely, a power series centered at a

Examples:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{10^k} (x - 5)^k$$

a power series centered at 5

$$\sum_{n=1}^{\infty} \frac{1}{n^2} x^n$$

a power series centered at 0

$$\sum_{k=0}^{\infty} k! (x - 1)^k$$

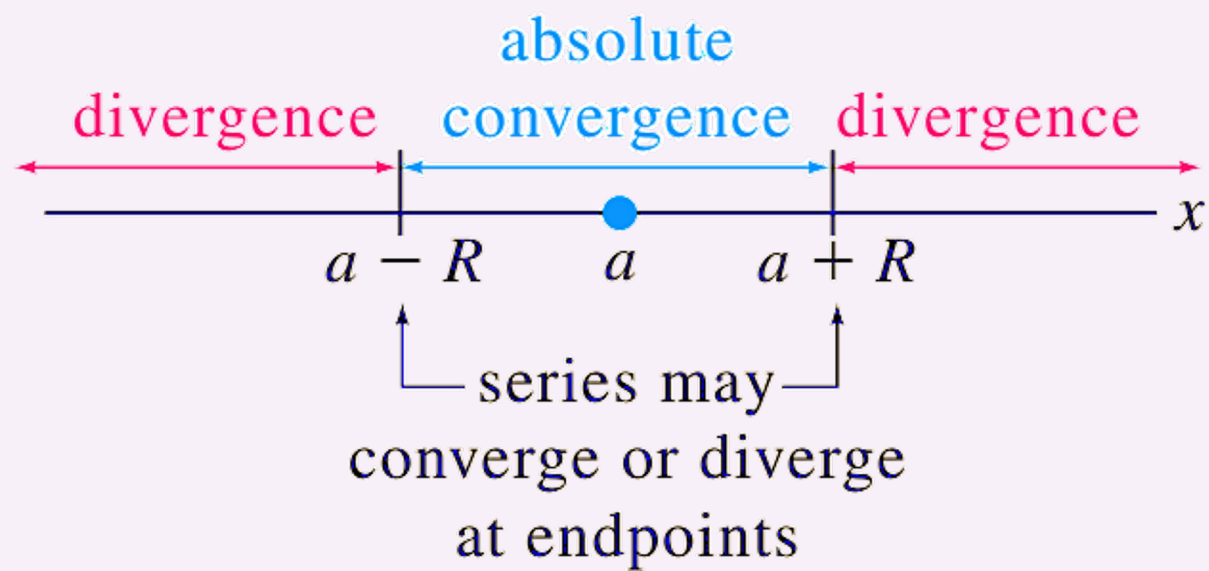
a power series centered at 1

Convergence: A power series is **convergent** at a specified value of x if its sequence of partial sums $\{S_N(x)\}$ converges, that is, $\lim_{N \rightarrow \infty} S_N(x) = \lim_{N \rightarrow \infty} \sum_{n=0}^N c_n(x - a)^n$ exists. If the limit does not exist at x , then the series is said to be **divergent**.

Interval of Convergence: Every power series has an **interval of convergence**. The interval of convergence is the set of all real numbers x for which the series converges. The center of the interval of convergence is the center a of the series.

Radius of Convergence: The radius R of the interval of convergence of a power series is called its **radius of convergence**. If $R > 0$, then a power series converges for $|x - a| < R$ and diverges for $|x - a| > R$. If the series converges only at its center a , then $R = 0$. If the series converges for all x , then we write $R = \infty$. Recall, the absolute-value inequality $|x - a| < R$ is equivalent to the simultaneous inequality $a - R < x < a + R$. A power series may or may not converge at the endpoints $a - R$ and $a + R$ of this interval.

Absolute Convergence: In the interior of its interval of convergence a power series **converges absolutely**. In other words, if x is in the interval of convergence and is not an endpoint of the interval, then the series of absolute values $\sum_{n=0}^{\infty} |c_n(x - a)^n|$ converges.



Ratio Test: Convergence of power series can often be determined by the **ratio test**.

Suppose $c_n \neq 0$ for all n in $\sum_{n=0}^{\infty} c_n(x - a)^n$, and that

$$\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| = |x - a| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = L.$$

If $L < 1$, the series converges absolutely; if $L > 1$ the series diverges; and if $L = 1$ the test is inconclusive. The ratio test is always inconclusive at an endpoint $a \pm R$.

Example. Find the interval and radius of convergence for $\sum_{n=1}^{\infty} \frac{(x-3)^n}{2^n n}$

Solution.

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(x-3)^{n+1}}{2^{n+1}(n+1)}}{\frac{(x-3)^n}{2^n n}} \right| = |x-3| \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} |x-3|$$



The series converges absolutely for

$$1 < x < 5$$

and $R = 2$

At endpoints:

$$x = 1 \quad \sum_{n=1}^{\infty} \frac{(1-3)^n}{2^n n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \quad \text{The series is convergent}$$

$$x = 5 \quad \sum_{n=1}^{\infty} \frac{(5-3)^n}{2^n n} = \sum_{n=1}^{\infty} \frac{2^n}{2^n n} = \sum_{n=1}^{\infty} \frac{1}{n} \quad \text{The series is divergent}$$



$[1, 5)$ is the interval of convergence of the series

Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n = f(a) + \frac{f'(a)}{1!} (x - a) + \frac{f''(a)}{1!} (x - a)^2 + \dots$$

Maclaurin series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{1!} x^2 + \dots$$

Also, recall that

Analytic at a Point: A function f is said to be **analytic at a point a** if it can be represented by a power series in $x - a$ with either a positive or an infinite radius of convergence.

Maclaurin Series	Interval of Convergence
$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$	$(-\infty, \infty)$
$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$	$(-\infty, \infty)$
$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$	$(-\infty, \infty)$
$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$	$[-1, 1] \quad (2)$
$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}$	$(-\infty, \infty)$
$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}$	$(-\infty, \infty)$
$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$	$(-1, 1]$
$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n$	$(-1, 1)$

Application

(for solving differential equations)

Assume

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

Here, a power series defines a function, whose domain is the interval of convergence of the series.

If $R > 0$ or $R = \infty$, $f(x)$ is continuous, differentiable, and integrable on the intervals

$$(-R, R) \text{ or } (-\infty, \infty)$$

$$y = \sum_{n=1}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots$$

$$y''' = \sum_{n=3}^{\infty} n(n-1)(n-2)c_n x^{n-3} = 6c_3 + 24c_4 x + 60c_5 x^2 + 120c_6 x^3 + \dots$$

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Now, for solving a differential equation like $y' + y = 0$, we can substitute a power series and find the coefficients accordingly

Before we use the power series method for solving ordinary differential equations, let's practice working with series

Exercises 6.1

In Problems 23 and 24 use a substitution to shift the summation index so that the general term of given power series involves x^k .

23.
$$\sum_{n=1}^{\infty} n c_n x^{n+2}$$

24.
$$\sum_{n=3}^{\infty} (2n - 1) c_n x^{n-3}$$

Solution:

$$\sum_{n=1}^{\infty} n c_n x^{n+2} = \sum_{k=3}^{\infty} (k-2) c_{k-2} x^k$$

✓ Define $k = n + 2 \rightarrow n = k - 2$

✓ Replace them

$$\sum_{n=3}^{\infty} (2n-1) c_n x^{n-3} = \sum_{k=0}^{\infty} (2(k+3)-1) c_{k+3} x^k = \sum_{k=0}^{\infty} (2k+5) c_{k+3} x^k$$

✓ Define $k = n - 3 \rightarrow n = k + 3$

✓ Replace them

In Problems 25–30 proceed as in Example 3 to rewrite the given expression using a single power series whose general term involves x^k .

$$25. \sum_{n=1}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

$$26. \sum_{n=1}^{\infty} n c_n x^{n-1} + 3 \sum_{n=0}^{\infty} c_n x^{n+2}$$

$$27. \sum_{n=1}^{\infty} 2n c_n x^{n-1} + \sum_{n=0}^{\infty} 6c_n x^{n+1}$$

$$28. \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+2}$$

$$29. \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - 2 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^n$$

$$30. \sum_{n=2}^{\infty} n(n-1) c_n x^n + 2 \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 3 \sum_{n=1}^{\infty} n c_n x^n$$

Solution:

$$\sum_{n=1}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n$$

✓ Define $k = n - 1 \rightarrow n = k + 1$

✓ Replace them



$$\sum_{k=0}^{\infty} (k + 1) c_{k+1} x^k - \sum_{k=0}^{\infty} c_k x^k = \sum_{k=0}^{\infty} [(k + 1) c_{k+1} - c_k] x^k$$

Solution:

$$\sum_{n=1}^{\infty} n c_n x^{n-1} + 3 \sum_{n=0}^{\infty} c_n x^{n+2}$$

✓ Define $k = n - 1 \rightarrow n = k + 1$

✓ Replace them

✓ Define $k = n + 2 \rightarrow n = k - 2$

✓ Replace them

$$\sum_{k=0}^{\infty} (k+1) c_{k+1} x^k + 3 \sum_{k=2}^{\infty} c_{k-2} x^k$$

for $k = 0, 1$, write it out of Σ

$$= c_1 + 2c_2x + \sum_{k=2}^{\infty} [(k+1)c_{k+1} + 3c_{k-2}] x^k$$

Solution:

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+2}$$

✓ Define $k = n - 2 \rightarrow n = k + 2$

✓ Replace them

✓ Define $k = n + 2 \rightarrow n = k - 2$

✓ Replace them

$$\sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k + \sum_{k=2}^{\infty} c_{k-2}x^k$$

for $k = 0, 1$, write it out of Σ

$$= 2c_2 + 6c_3x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + c_{k-2}]x^k$$

In Problems 31–34 verify by direct substitution that the given power series is a solution of the indicated differential equation. [*Hint*: For a power x^{2n+1} let $k = n + 1$.]

$$31. y = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n}, \quad y' + 2xy = 0$$

$$32. y = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad (1 + x^2)y' + 2xy = 0$$

$$33. y = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n, \quad (x + 1)y'' + y' = 0$$

$$34. y = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n}(n!)^2} x^{2n}, \quad xy'' + y' + xy = 0$$

Solution:

$$31. y = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n}, \quad y' + 2xy = 0$$

Step 1. find $2xy$ and y'

$$2xy = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2}{n!} x^{2n+1}$$

$$y' = \sum_{n=0}^{\infty} \frac{(-1)^n}{\cancel{n!}} \cdot \cancel{2n} x^{2n-1}$$

$(n-1)!$

Define $k = n$

Step 2. make them compatible

Define $n = k + 1 \rightarrow 2n - 1 = 2k + 1$

$$y' = \sum_{k=0}^{\infty} \frac{(-1)^{k+1} \cdot 2}{k!} x^{2k+1} = - \sum_{k=0}^{\infty} \frac{(-1)^k \cdot 2}{k!} x^{2k+1} \qquad 2xy = \sum_{k=0}^{\infty} \frac{(-1)^k \cdot 2}{k!} x^{2k+1}$$

Step 3. replace in DE

$$y' + 2xy = - \sum_{k=0}^{\infty} \frac{(-1)^k \cdot 2}{k!} x^{2k+1} + \sum_{k=0}^{\infty} \frac{(-1)^k \cdot 2}{k!} x^{2k+1} = 0$$

32. $y = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad (1 + x^2)y' + 2xy = 0$

Solution:

Step 1. find $2xy$, y' and x^2y'

$$y' = \sum_{n=0}^{\infty} (-1)^n \cdot 2n x^{2n-1}$$

$$x^2y' = \sum_{n=1}^{\infty} (-1)^n \cdot 2n x^{2n+1}$$

$$2xy = 2x \sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} (-1)^n \cdot 2 x^{2n+1}$$

Step 2. make them compatible

$$\checkmark \quad y' = \sum_{n=1}^{\infty} (-1)^n \cdot 2n x^{2n-1}$$

Define $n = k + 1 \rightarrow 2n - 1 = 2k + 1$

$$\begin{aligned} y' &= \sum_{k=0}^{\infty} (-1)^{k+1} \cdot 2(k+1) x^{2k+1} = - \sum_{k=0}^{\infty} (-1)^k \cdot 2(k+1) x^{2k+1} \\ &= -2x - \sum_{k=1}^{\infty} (-1)^k \cdot 2(k+1) x^{2k+1} \end{aligned}$$

$$\checkmark \quad x^2 y' = \sum_{n=1}^{\infty} (-1)^n \cdot 2n x^{2n+1} \quad \text{Define } k = n: \quad x^2 y' = \sum_{k=1}^{\infty} (-1)^k \cdot 2k x^{2k+1}$$

$$\begin{aligned} \checkmark \quad 2xy &= \sum_{n=0}^{\infty} (-1)^n \cdot 2 x^{2n+1} \quad \text{Define } k = n: \quad 2xy = \sum_{k=0}^{\infty} (-1)^k \cdot 2 x^{2k+1} \\ &= 2x + \sum_{k=1}^{\infty} (-1)^k \cdot 2 x^{2k+1} \end{aligned}$$

Step 3. replace in DE

$$\begin{aligned}y' + x^2 y' + 2xy &= \left(-2x - \sum_{k=1}^{\infty} (-1)^k \cdot 2(k+1)x^{2k+1} \right) + \left(\sum_{k=1}^{\infty} (-1)^k \cdot 2kx^{2k+1} \right) + \left(2x + \sum_{k=1}^{\infty} (-1)^k \cdot 2x^{2k+1} \right) \\&= (-2x + 2x) + \sum_{k=1}^{\infty} (-1)^k \cdot 2[-(k+1) + k+1] x^{2k+1} \\&= 0 + \sum_{k=1}^{\infty} (-1)^k \cdot 2(0)x^{2k+1} \\&= 0\end{aligned}$$

Example. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$y' + y = 0$$

Solution :

Step 1. Find y'

$$y = \sum_{n=0}^{\infty} c_n x^n \xrightarrow{\text{Find the derivative}} y' = \sum_{n=1}^{\infty} c_n n x^{n-1}$$

Step2. make them compatible

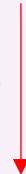
$$\begin{array}{ccc} y = \sum_{n=0}^{\infty} c_n x^n & & y' = \sum_{n=1}^{\infty} c_n n x^{n-1} \\ \text{Define } k = n \downarrow & & \downarrow \text{Define } k = n - 1 \rightarrow n = k + 1 \\ y = \sum_{k=0}^{\infty} c_k x^k & & y' = \sum_{k=0}^{\infty} (k + 1) c_{k+1} x^k \end{array}$$

Step 3. substitute the series in $y' + y = 0$

$$\sum_{k=0}^{\infty} (k+1)c_{k+1}x^k + \sum_{k=0}^{\infty} c_k x^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)c_{k+1} + c_k] x^k = 0$$

By identity property $c_{k+1}(k+1) + c_k = 0$



Step4. find the coefficients $c_{k+1} = -\frac{1}{k+1} c_k, \quad k = 0, 1, 2, \dots$

$$c_1 = -\frac{1}{1}c_0 = -c_0$$

$$c_2 = -\frac{1}{2}c_1 = -\frac{1}{2}(-c_0) = \frac{1}{2}c_0$$

$$c_3 = -\frac{1}{3}c_2 = -\frac{1}{3}\left(\frac{1}{2}c_0\right) = -\frac{1}{3 \cdot 2}c_0$$

$$c_4 = -\frac{1}{4}c_3 = -\frac{1}{4}\left(-\frac{1}{3 \cdot 2}c_0\right) = \frac{1}{4 \cdot 3 \cdot 2}c_0$$



Step 5. obtain the power series solution

$$\begin{aligned}y &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots \\&= c_0 - c_0x + \frac{1}{2}c_0x^2 - c_0\frac{1}{3 \cdot 2}x^3 + c_0\frac{1}{4 \cdot 3 \cdot 2}x^4 - \dots \\&= c_0\left[1 - x + \frac{1}{2}x^2 - \frac{1}{3 \cdot 2}x^3 + \frac{1}{4 \cdot 3 \cdot 2}x^4 - \dots\right].\end{aligned}$$



$$y = c_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^k, \quad \text{The interval of convergence is } (-\infty, \infty)$$

or $y = c_0 e^{-x}$

Note that we used identity property of power series

Identity Property: If $\sum_{n=0}^{\infty} c_n(x-a)^n = 0$, $R > 0$, for all numbers x in some open interval, then $c_n = 0$ for all n .

Exercise 35. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$y' - 5y = 0$$

Solution (similarly):

Step 1. Find y'

$$y = \sum_{n=0}^{\infty} c_n x^n \xrightarrow{\text{Find the derivative}} y' = \sum_{n=1}^{\infty} c_n n x^{n-1}$$

Step 2. make them compatible

$$\begin{array}{ccc} y = \sum_{n=0}^{\infty} c_n x^n & & y' = \sum_{n=1}^{\infty} c_n n x^{n-1} \\ \text{Define } k = n \downarrow & & \downarrow \text{Define } k = n - 1 \rightarrow n = k + 1 \\ y = \sum_{k=0}^{\infty} c_k x^k & & y' = \sum_{k=0}^{\infty} (k + 1) c_{k+1} x^k \end{array}$$

Step 3. substitute the series in $y' - 5y = 0$

$$\sum_{k=0}^{\infty} (k+1)c_{k+1}x^k - \sum_{k=0}^{\infty} 5c_kx^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)c_{k+1} - 5c_k] x^k = 0$$

$$= 0$$

Step 4. find the coefficients $c_{k+1} = \frac{5}{k+1} c_k \quad k = 0, 1, 2, \dots$



$$\begin{aligned} c_1 &= 5c_0 \\ c_2 &= \frac{5}{2}c_1 = \frac{5^2}{2!}c_0 \\ c_3 &= \frac{5}{3}c_2 = \frac{5^3}{3!}c_0 \\ &\vdots \\ c_n &= \frac{5^n}{n!}c_0 \end{aligned}$$

Step 5. obtain the power series solution

$$y = c_0 \sum_{n=0}^{\infty} \frac{5^n}{n!} x^n = c_0 \sum_{n=0}^{\infty} \frac{(5x)^n}{n!} = c_0 e^{5x}$$

Exercise 37. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$y' = xy$$

Solution (similarly):

Step 1. Find y' and xy :

$$y = \sum_{n=0}^{\infty} c_n x^n \quad \begin{array}{l} \rightarrow y' = \sum_{n=1}^{\infty} c_n n x^{n-1} \\ \rightarrow xy = x \cdot \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} c_n x^{n+1} \end{array}$$

Step 2. make them compatible

$xy = \sum_{n=0}^{\infty} c_n x^{n+1}$	$y' = \sum_{n=1}^{\infty} c_n n x^{n-1}$
Define $k = n + 1 \rightarrow n = k - 1$	Define $k = n - 1 \rightarrow n = k + 1$
$xy = \sum_{k=1}^{\infty} c_{k-1} x^k$	$y' = \sum_{k=0}^{\infty} (k + 1) c_{k+1} x^k$
	$= c_1 + \sum_{k=1}^{\infty} (k + 1) c_{k+1} x^k$

Step 3. substitute the series in $y' - xy = 0$

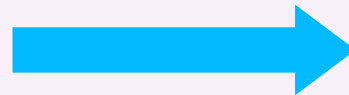
$$\begin{aligned} & \sum_{k=0}^{\infty} (k+1)c_{k+1}x^k - \sum_{k=1}^{\infty} c_{k-1}x^k \\ &= c_1 + \sum_{k=1}^{\infty} [(k+1)c_{k+1} - c_{k-1}]x^k \\ c_1 = 0 & \leftarrow \end{aligned}$$

$= 0$

$$(k+1)c_{k+1} - c_{k-1} = 0$$

Step 4. find the coefficients

$$c_{k+1} = \frac{c_{k-1}}{k+1}$$



$$\begin{aligned} c_1 &= 0 \\ c_2 &= \frac{c_0}{2} \\ c_3 &= 0 \\ c_4 &= \frac{c_0}{2 \cdot 4} = \frac{c_0}{2^2 \cdot 2!} \\ c_5 &= 0 \\ c_6 &= \frac{c_0}{2^3 \cdot 3!} \\ &\vdots \end{aligned}$$

$$c_1 = 0$$

$$c_2 = \frac{c_0}{2}$$

$$c_3 = 0$$

$$c_4 = \frac{c_0}{2 \cdot 4} = \frac{c_0}{2^2 \cdot 2!}$$

$$c_5 = 0$$

$$c_6 = \frac{c_0}{2^3 \cdot 3!}$$

$$c_7 = 0$$

$$c_8 = \frac{c_0}{2^4 \cdot 4!}$$

$\vdots$

$$c_n = \begin{cases} \frac{c_0}{2^m \cdot m!} & \text{if } n = 2m \ (m = 0, 1, 2, \dots), \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$



Step 5. obtain the power series solution

$$y = c_0 \sum_{m=0}^{\infty} \frac{1}{2^m \cdot m!} x^{2m} = c_0 \sum_{m=0}^{\infty} \frac{\left(\frac{x^2}{2}\right)^m}{m!}$$

$$y = c_0 e^{\frac{x^2}{2}}$$

Exercise 36. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$4y' + y = 0$$

Exercise 38. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$(1 + x)y' + y = 0$$

Second-order DE

$$y'' + P(x)y' + Q(x)y = 0.$$

Definition

A point $x = x_0$ is said to be an **ordinary point** of the differential of the differential equation

$$y'' + P(x)y' + Q(x)y = 0$$

if both coefficients $P(x)$ and $Q(x)$ are analytic at x_0 . A point that is not an ordinary point of the DE, is said to be a **singular point** of the DE.

Examples

$$y'' + y = 0$$

$$y'' + 3y' + 2y = 0,$$

$$y'' + e^x y' + (\sin x)y = 0.$$

These DEs have no singular points (every finite value of x is an ordinary point of the differential equations)

Examples

$$y'' + xy' + (\ln x)y = 0$$

$P(x)$ is analytic at every real number,

$Q(x) = \ln(x)$ is analytic at every positive real number; Also, it is discontinuous at $x = 0$,

thus $x = 0$ is a singular point of the DE

$$y'' + \frac{1}{x}y' + y = 0,$$

$Q(x)$ is analytic at every real number,

$P(x) = \frac{1}{x}$ is discontinuous at $x = 0$, thus $x = 0$ is a singular point of the DE

Examples

$$(x^2 - 1)y'' + 2xy' + 6y = 0$$

$P(x) = \frac{2x}{x^2-1}$, and $Q(x) = \frac{1}{x^2-1}$ are discontinuous at $x = \pm 1$,

thus $x = \pm 1$ are singular points of the DE

$$(x^2 + 1)y'' + xy' - y = 0$$

$P(x) = \frac{x}{x^2+1}$, and $Q(x) = \frac{-1}{x^2+1}$ are discontinuous at $x = \pm i$,

thus $x = \pm i$ are singular points of the DE

Theorem(Existence of Power Series Solutions)

If $x = x_0$ is an ordinary point of the differential equation

$$y'' + P(x)y' + Q(x)y = 0$$

we can always find two linearly independent solutions in the form of a power series centered at x_0 , that is,

$$y = \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

A power series solution converges at least on some interval defined by $|x - x_0| < R$ where R is the distance from x_0 to the closest singular point.

Examples: Find the minimum radius of convergence of a power series solution of the second-order DE

$$(x^2 - 2x + 5)y'' + xy' - y = 0$$

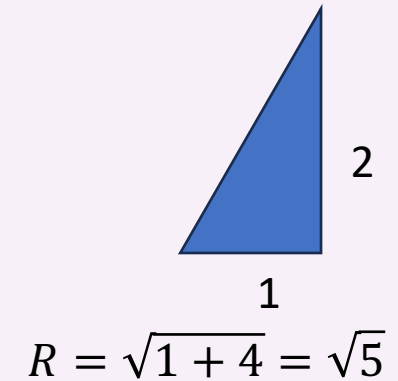
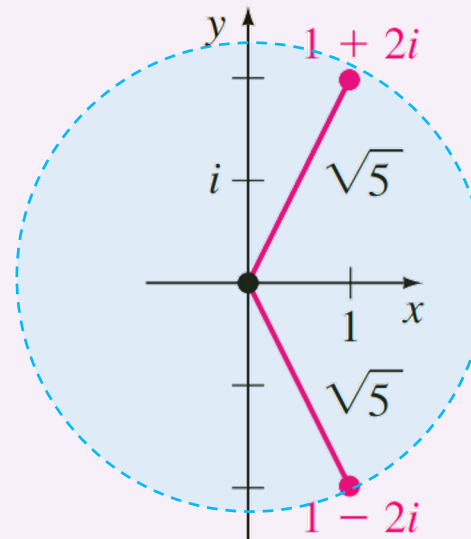
- about the ordinary point $x = 0$
- about the ordinary point $x = -1$

Solution :

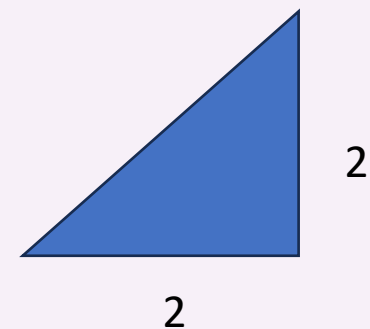
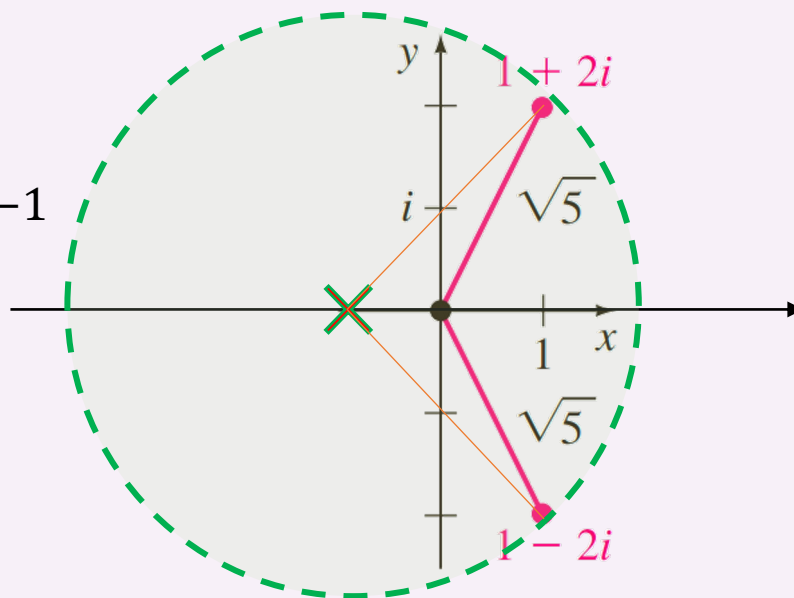
$P(x)$ and $Q(x)$ are discontinuous at $x = 1 \pm 2i$, thus $x = 1 \pm 2i$, are singular points of the DE

- about the ordinary point $x = 0$

❖ The solution $y = \sum_{n=0}^{\infty} c_n x^n$ is convergent in this area (blue disk)



- about the ordinary point $x = -1$



$$R = \sqrt{4 + 4} = \sqrt{8}$$

❖ The solution $y = \sum_{n=0}^{\infty} c_n (x + 1)^n$ is convergent in this area (green disk)

- about the ordinary point $x = +1$: $R = 2$

Example. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$y'' - xy = 0$$

The differential equation doesn't have any singular point.

According to the theorem, two power series solutions, centered at $x = 0$, converges for $|x| < \infty$

Solution

$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \xrightarrow{n-2=k} y'' = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k$$

and

$$xy = \sum_{n=0}^{\infty} c_n x^{n+1} \xrightarrow{n+1=k} xy = \sum_{k=1}^{\infty} c_{k-1} x^k$$

$$\Rightarrow y'' - xy = 2c_2 + \sum_{k=1}^{\infty} [(k+1)(k+2)c_{k+2} - c_{k-1}]x^k = 0.$$

$$\Rightarrow c_2 = 0 \text{ and } (k+1)(k+2)c_{k+2} - c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

$$\Rightarrow c_{k+2} = \frac{c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \dots$$

$$\Rightarrow k = 1, \quad c_3 = \frac{c_0}{2 \cdot 3}$$

$$k = 2, \quad c_4 = \frac{c_1}{3 \cdot 4}$$

$$k = 3, \quad c_5 = \frac{c_2}{4 \cdot 5} = 0$$

$$k = 4, \quad c_6 = \frac{c_3}{5 \cdot 6} = \frac{1}{2 \cdot 3 \cdot 5 \cdot 6} c_0$$

$$k = 5, \quad c_7 = \frac{c_4}{6 \cdot 7} = \frac{1}{3 \cdot 4 \cdot 6 \cdot 7} c_1$$

$$k = 6, \quad c_8 = \frac{c_5}{7 \cdot 8} = 0$$

$$k = 7, \quad c_9 = \frac{c_6}{8 \cdot 9} = \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9} c_0$$

$$k = 8, \quad c_{10} = \frac{c_7}{9 \cdot 10} = \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10} c_1$$

$$k = 9, \quad c_{11} = \frac{c_8}{10 \cdot 11} = 0$$

substitutue the obtained cofficients in the power series solution

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + c_{11}x^{11} + \dots$$

$$y = c_0 + c_1x + 0 + \frac{c_0}{2 \cdot 3}x^3 + \frac{c_1}{3 \cdot 4}x^4 + 0 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6}x^6 \\ + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7}x^7 + 0 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9}x^9 + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10}x^{10} + 0 + \dots = c_0y_1(x) + c_1y_2(x)$$

where


$$y_1(x) = 1 + \frac{1}{2 \cdot 3}x^3 + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6}x^6 + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9}x^9 + \dots = 1 + \sum_{k=1}^{\infty} \frac{1}{2 \cdot 3 \cdots (3k-1)(3k)}x^{3k} \\ y_2(x) = x + \frac{1}{3 \cdot 4}x^4 + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7}x^7 + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10}x^{10} + \dots = x + \sum_{k=1}^{\infty} \frac{1}{3 \cdot 4 \cdots (3k)(3k+1)}x^{3k+1}$$

Example. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

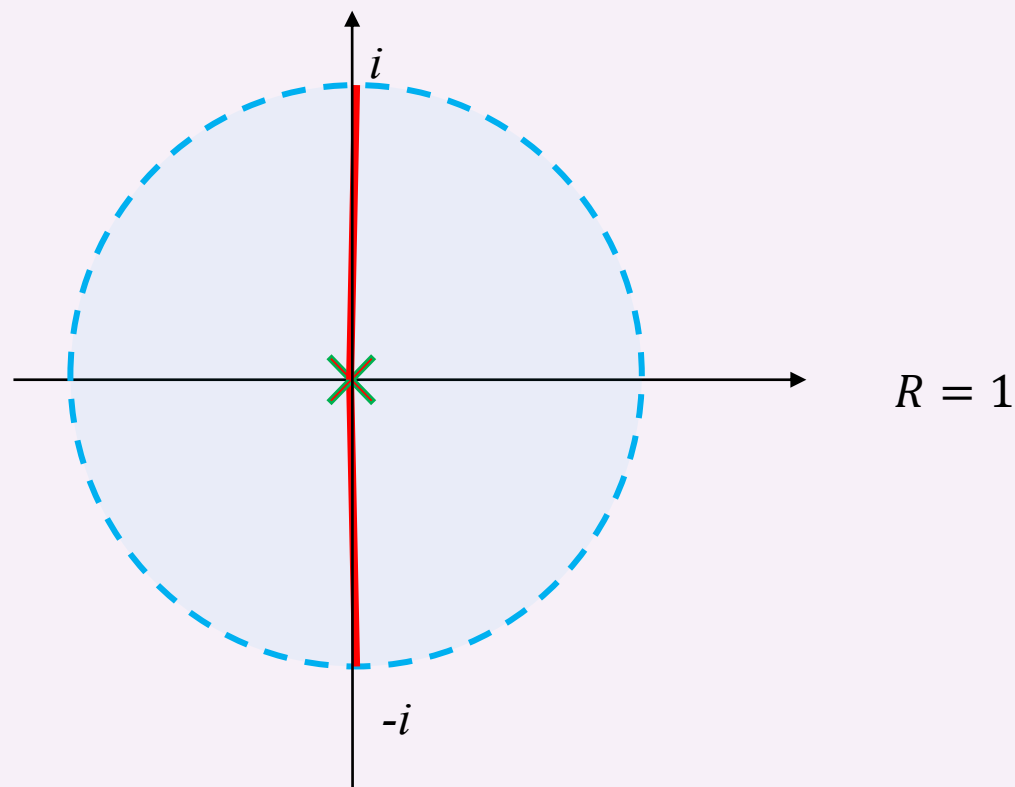
$$(x^2 + 1)y'' + xy' - y = 0$$

$P(x) = \frac{x}{x^2+1}$, and $Q(x) = \frac{-1}{x^2+1}$ are discontinuous at $x = \pm i$,

The differential equation has two singular points $x = \pm i$.

According to the theorem, two power series solutions, centered at $x = 0$, converges for $|x| < 1$

about the ordinary point $x = 0$



❖ The solution $y = \sum_{n=0}^{\infty} c_n x^n$ is convergent in this area (blue disk)

$$(x^2 + 1)y'' + xy' - y = 0$$

Solution

$$y = \sum_{n=0}^{\infty} c_n x^n \xrightarrow{n=k} y = \sum_{k=0}^{\infty} c_k x^k$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \xrightarrow{n-2=k} y'' = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k$$

and

$$x^2 y'' = x^2 \cdot \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} = \sum_{n=2}^{\infty} n(n-1)c_n x^n \xrightarrow{n=k} x^2 y'' = \sum_{k=2}^{\infty} k(k-1)c_k x^k$$

$$xy' = x \cdot \sum_{n=1}^{\infty} n c_n x^{n-1} = \sum_{n=1}^{\infty} n c_n x^n \xrightarrow{n=k} xy' = \sum_{k=1}^{\infty} k c_k x^k$$

$$\underbrace{\sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k + \sum_{k=2}^{\infty} k(k-1)c_k x^k}_{\text{from } (x^2+1)y''} + \underbrace{\sum_{k=1}^{\infty} k c_k x^k}_{\text{from } xy'} - \underbrace{\sum_{k=0}^{\infty} c_k x^k}_{\text{from } -y} = 0$$

➡ $2c_2 - c_0 + 6c_3x + \sum_{k=2}^{\infty} [k(k-1)c_k + (k+2)(k+1)c_{k+2} + kc_k - c_k] x^k = 0$

So, $\begin{cases} 2c_2 - c_0 = 0 \\ c_3 = 0 \\ (k+1)(k-1)c_k + (k+2)(k+1)c_{k+2} = 0 \quad (k \geq 2) \end{cases}$ ➡ $\begin{cases} c_2 = \frac{c_0}{2} \\ c_3 = 0 \\ c_{k+2} = -\frac{(k-1)}{k+2}c_k \quad (k \geq 2) \end{cases}$

For $k = 2, 3, \dots$

$$c_4 = -\frac{1}{4}c_2 = -\frac{1}{2 \cdot 4}c_0 = -\frac{1}{2^2 2!}c_0$$

$$c_5 = -\frac{2}{5}c_3 = 0 \quad \leftarrow c_3 \text{ is zero}$$

$$c_6 = -\frac{3}{6}c_4 = \frac{3}{2 \cdot 4 \cdot 6}c_0 = \frac{1 \cdot 3}{2^3 3!}c_0$$

$$c_7 = -\frac{4}{7}c_5 = 0 \quad \leftarrow c_5 \text{ is zero}$$

$$c_8 = -\frac{5}{8}c_6 = -\frac{3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 8}c_0 = -\frac{1 \cdot 3 \cdot 5}{2^4 4!}c_0$$

$$c_9 = -\frac{6}{9}c_7 = 0, \quad \leftarrow c_7 \text{ is zero}$$

$$c_{10} = -\frac{7}{10}c_8 = \frac{3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10}c_0 = \frac{1 \cdot 3 \cdot 5 \cdot 7}{2^5 5!}c_0,$$

substitutue the obtained cofficients in the power series solution

$$\begin{aligned}y &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + \dots \\&= c_0 \left[1 + \frac{1}{2}x^2 - \frac{1}{2^2 2!}x^4 + \frac{1 \cdot 3}{2^3 3!}x^6 - \frac{1 \cdot 3 \cdot 5}{2^4 4!}x^8 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2^5 5!}x^{10} - \dots \right] + c_1x \\&= c_0 y_1(x) + c_1 y_2(x).\end{aligned}$$

where

$$\begin{cases} y_1(x) = 1 + \frac{1}{2}x^2 - \frac{1}{2^2 2!}x^4 + \frac{1 \cdot 3}{2^3 3!}x^6 - \frac{1 \cdot 3 \cdot 5}{2^4 4!}x^8 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2^5 5!}x^{10} - \dots \\ y_2(x) = x \end{cases}$$

Example. For finding power series solutions $y = \sum_{n=0}^{\infty} c_n x^n$ of the differential equation

$$y'' - (1 + x)y = 0,$$

we obtained $c_2 = \frac{1}{2}c_0$, and

$$c_{k+2} = \frac{c_k + c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \dots$$

Find y_1 and y_2

The differential equation has no singular point

Solution: Using $c_2 = \frac{1}{2}c_0$, and the recurrence relation,

$$c_3 = \frac{c_1 + c_0}{3 \times 2} = \frac{c_1 + c_0}{6}$$

$$c_4 = \frac{\frac{1}{2}c_0 + c_1}{4 \times 3} = \frac{c_1 + \frac{1}{2}c_0}{12}$$

$$c_5 = \frac{\frac{c_1 + c_0}{6} + \frac{1}{2}c_0}{5 \times 4} = \frac{\frac{c_1 + c_0 + 3c_0}{6}}{20} = \frac{c_1 + 4c_0}{120}$$

$$c_6 = \frac{\frac{c_1 + \frac{1}{2}c_0}{12} + \frac{c_1 + c_0}{6}}{6 \times 5} = \frac{\frac{c_1 + \frac{1}{2}c_0 + 2c_1 + 2c_0}{12}}{30} = \frac{3c_1 + \frac{5}{2}c_0}{360} = \frac{6c_1 + 5c_0}{720}$$

➡ $y = c_0 + c_1x + \frac{1}{2}c_0x^2 + \frac{c_1 + c_0}{6}x^3 + \frac{c_1 + \frac{1}{2}c_0}{12}x^4 + \frac{c_1 + 4c_0}{120}x^5 + \frac{6c_1 + 5c_0}{720}x^6 + \dots$

However it can be shown as a linear combination of two linearly independent solutions,

$$c_0 y_1(x) + c_1 y_2(x)$$

but the calculation is too frustrating. For these cases, when in the recurrence relation each term depends on two previous terms, like here,

$$c_{k+2} = \frac{c_k + c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \dots$$

Instead of directly finding $y(x) = c_0 y_1(x) + c_1 y_2(x)$, we find y_1 and y_2 separately

To do this, once let $c_0 = 0, c_1 = 1$ and then $c_1 = 0, c_0 = 1$

Now let's repeat the solution

$$c_1 = 0, c_0 = 1$$

$$c_2 = \frac{1}{2} c_0 = \frac{1}{2}$$

$$c_3 = \frac{c_1 + c_0}{2 \cdot 3} = \frac{0 + 1}{6} = \frac{1}{6}$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{\frac{1}{2} + 0}{12} = \frac{1}{24}$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{\frac{1}{6} + \frac{1}{2}}{20} = \frac{1}{30}$$

⋮

$$c_0 = 0, c_1 = 1$$

$$c_2 = \frac{1}{2} c_0 = 0$$

$$c_3 = \frac{c_1 + c_0}{2 \cdot 3} = \frac{1 + 0}{6} = \frac{1}{6}$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{0 + 1}{12} = \frac{1}{12}$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{\frac{1}{6} + 0}{20} = \frac{1}{120}$$

⋮



$$y_1(x) = 1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{30}x^5 + \dots$$

$$y_2(x) = x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{120}x^5 + \dots$$

Exercises 6.2

In Problems 7–18 find two power series solutions of the given differential equation about the ordinary point $x = 0$.

7. $y'' + xy = 0$

8. $y'' + x^2y = 0$

9. $y'' - 2xy' + y = 0$

10. $y'' - xy' + 2y = 0$

11. $y'' + x^2y' + xy = 0$

12. $y'' + 2xy' + 2y = 0$

13. $(x - 1)y'' + y' = 0$

14. $(x + 2)y'' + xy' - y = 0$

15. $y'' - (x + 1)y' - y = 0$

16. $(x^2 + 1)y'' - 6y = 0$

17. $(x^2 + 2)y'' + 3xy' - y = 0$

18. $(x^2 - 1)y'' + xy' - y = 0$